FORM TP 2013002

TEST CODE **01207020**

JANUARY 2013

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®

EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question, in this booklet.
4. Where appropriate, answers should be illustrated by diagrams.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. An athlete who normally lives and trains at sea level has maintained a red blood cell count of 4.2 million per unit of blood. In preparation for an athletic event scheduled to take place 2,500 m above sea level, he began training at this altitude six months prior to the event. His number of red blood cells was counted at monthly intervals. Table 1 shows the values of the red blood cell count for the six-month period.

TABLE 1: ATHLETE'S RED BLOOD CELL COUNT DURING SIX MONTHS OF TRAINING AT 2,500 m ABOVE SEA LEVEL

Month	Number of Red Blood Cells (millions/unit of blood)
1	4.4
2	4.7
3	5.1
4	5.2
5	5.3
6	5.3

(a) (i) **On the grid provided below**, draw a graph to represent the data in Table 1.

(5 marks)

(ii) Describe the shape of the graph between Months 1 and 3 and between Months 4 and 6.

Months 1–3 ______________________________

____

Months 4–6 ______________________________

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(2 marks)

(iii) Taking into account the concentration of oxygen at higher altitudes, give an explanation for the change in the red blood cell count at higher altitudes.

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(2 marks)

(iv) Suggest a suitable hypothesis for the results shown in Table 1.

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(2 marks)

(v) What would be a suitable control in an experiment being conducted to test the hypothesis suggested in (iv) above?

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(2 marks)

(b) Katz is suffering from sickle-cell anaemia.

(i) How do the red blood cells of Katz differ from normal red blood cells?

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(2 marks)

(ii) Explain why sickle-cell anaemia makes Katz feel tired easily.

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(2 marks)

(c) (i) Suggest ONE condition under which the number of **white** blood cells in the human body will be increased.

____

(1 mark)

(ii) State the function of TWO different types of white blood cells.

____

____

(2 marks)

(iii) Suggest why red blood cells have a shorter life span than that of white blood cells.

____

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(1 mark)

(d) Plants differ from animals in the way that they protect themselves and in their mode of nutrition.

(i) Describe TWO ways by which plants protect themselves from excess water loss.

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(2 marks)

(ii) State the name used to describe the mode of nutrition in plants.

____

(1 mark)

(iii) Name the organelle found in the cells of green plants that is responsible for the type of nutrition stated in (ii) above.

____

(1 mark)

Total 25 marks

2. Figure 1 is a diagram of a tomato plant.

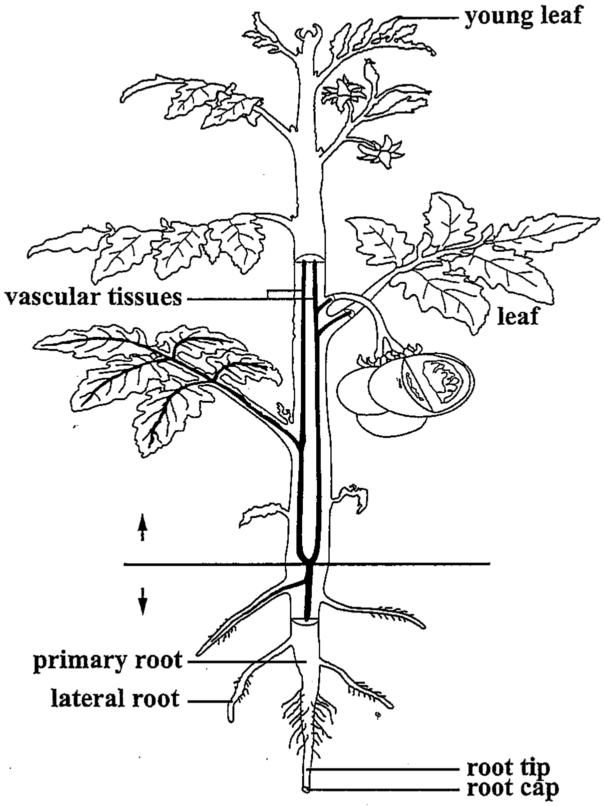

Figure 1. Diagram of a tomato plant

(a) Name the part of the plant in Figure 1 that is responsible for EACH of the following functions:

(i) Support of the stem ______________________ **(1 mark)**

(ii) Sexual reproduction ______________________ **(1 mark)**

(iii) Development of the embryo ______________________ **(1 mark)**

(iv) Obtaining water and mineral salts ______________________ **(1 mark)**

(v) Making food for the plant ______________________ **(1 mark)**

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(b) (i) State the method of fruit dispersal for the plant in Figure 1, and give ONE reason to support your answer.

Method: ______________________________

Reason: ______________________________

(2 marks)

(ii) Explain why plants produced from the seeds of the tomato plant are NOT identical to the parent plant.

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(2 marks)

(c) (i) Suggest TWO advantages that may be gained by propagating this plant asexually, for example, by tissue culture.

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(2 marks)

(ii) Suggest TWO characteristics that a farmer may want to keep in his crop of tomato plants.

____

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(2 marks)

(d) Genetic engineering has been used to modify tomato plants.

Outline ONE concern that has been raised about the consumption of genetically modified foods.

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____

(2 marks)

Total 15 marks

3. (a) Figure 2 shows an outline of part of the human body.

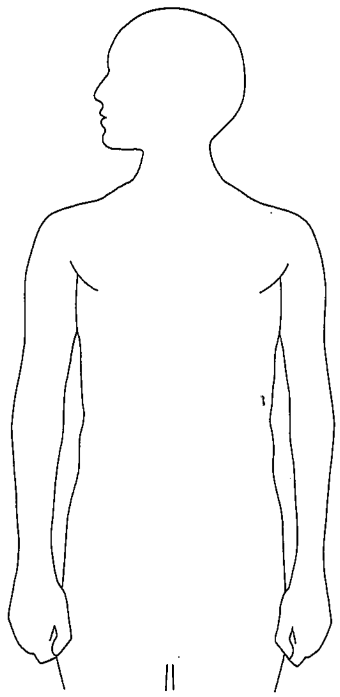

Figure 2. Outline of part of the human body

Use a label line in Figure 2 to illustrate the position of EACH of the following endocrine glands:

(i) Adrenals **(1 mark)**

(ii) Thyroid gland **(1 mark)**

(iii) The pituitary gland **(1 mark)**

(b) The pituitary gland produces several hormones that affect the function of other glands.

(i) Name TWO hormones produced by the pituitary gland and state the role of EACH.

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(4 marks)

(ii) Suggest TWO effects of a poorly developed pituitary gland in humans.

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(2 marks)

(c) Plants also produce hormones in the tip of their shoots that help them to respond to external stimuli such as light. Figure 3 illustrates an experiment to investigate a shoot's response to light.

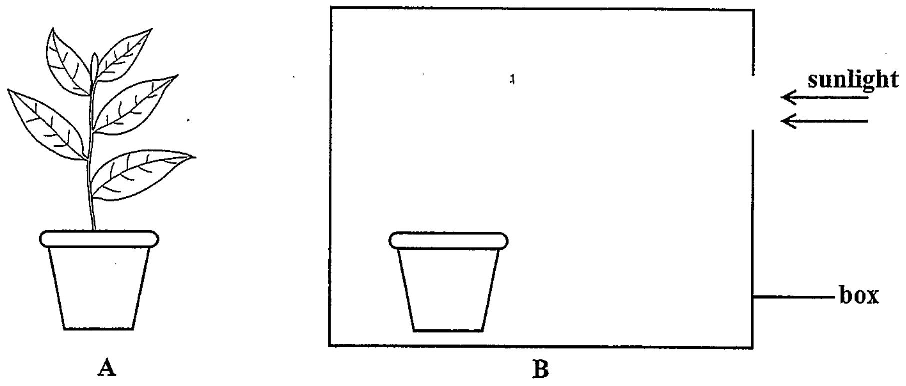

Figure 3. An experiment involving a shoot

(i) Figure 3A shows a shoot growing in sunlight. Complete Figure 3B to show the

a) response of the shoot to light coming from **only one** direction **(1 mark)**

b) distribution of the hormone responsible for the response shown in your drawing in B. **(1 mark)**

(ii) Explain how the hormone causes the response shown in Figure 3B.

____

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(2 marks)

(iii) Suggest why this response to light is important to the survival of the plant.

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(2 marks)

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) (i) Name the structures responsible for gaseous exchange in humans and in fish. **(2 marks)**

(ii) Identify THREE features that the structures named in (i) above have in common. **(3 marks)**

(b) Gaseous exchange occurs in plants as well as in animals.

(i) Draw and annotate a diagram to show how oxygen and carbon dioxide are exchanged across the respiratory surface of a human. **(3 marks)**

(ii) a) Name the process by which gases are exchanged in plants. **(1 mark)**

b) Explain why in plants there is a difference in the percentage of the respiratory gases exchanged during the day compared with the percentage exchanged during the night. **(3 marks)**

(c) Explain the effect of cutting and burning forests on the operation of the carbon cycle, and on the environment. **(3 marks)**

Total 15 marks

Write your answer to Question 4 here.

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Space for diagram

5. (a) (i) With the aid of a diagram, describe how a nerve cell is adapted to carry out its function. **(6 marks)**

(ii) Suggest how loss of nerve cells from the brain may lead to loss of memory. **(2 marks)**

(b) Discuss the abuse of alcohol in humans. In your answer, identify THREE effects caused by the action of alcohol on the brain, and suggest TWO **social** and TWO **economic** consequences of alcohol abuse on society. **(7 marks)**

Total 15 marks

Space for diagram

Write your answer to Question 5 here.

Write your answer to Question 5 here.

6. (a) A recent outbreak of swine flu was caused by the H1N1 virus. One prevention measure adopted by some countries was an immunization programme.

(i) Distinguish between artificial immunity and natural immunity. In your answer, explain how artificial immunity and natural immunity protect the human body against infectious diseases. **(6 marks)**

(ii) Suggest TWO **disadvantages** of immunization. **(2 marks)**

(iii) The H1N1 virus is spread through the air. Suggest TWO precautions, other than immunization, that an individual can take to avoid getting the disease. **(2 marks)**

(b) (i) Name a pathogenic disease that is spread in a manner different from the H1N1 virus and suggest ONE measure that may be taken to control the spread of this disease. **(2 marks)**

(ii) How are the methods used in the control of the spread of viruses among crop plants similar to those used to control the spread of viruses among farm animals? **(3 marks)**

Total 15 marks

Write your answer to Question 6 here.

Write your answer to Question 6 here.

Write your answer to Question 6 here.

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.